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To
The Editor-in-Chief
International Journal of Communication Systems
Wiley

Subject: Submission of original research manuscript for consideration

Dear Editor-in-Chief,

Please find enclosed our research manuscript titled **“Predictive Temporal Bridge: AI Architecture for Deep-Space Command Synchronization,”** which we are submitting for consideration as an original research article for publication in the *International Journal of Communication Systems*.

This study addresses a critical operational and psychological bottleneck in interplanetary human collaboration: the severe round-trip light-time delay (8 to 45 minutes at Mars distances) that disrupts conversational grounding and isolates crews. Traditional space systems isolate the crew in an “autonomy bubble,” leading to psychological reactance and relational friction with Ground Control. We propose the **Predictive Temporal Bridge (PTB)**, an artificial intelligence-mediated communication (AI-MC) interface that reconstructs a virtual present using dual-AI state integration, thereby sustaining social presence and shared situational awareness (SA).

We evaluate this framework using a high-fidelity interactive simulation of a critical Mars orbital insertion burn anomaly. We trace the subjective and objective effects on communication dynamics using four key indicators: the Operator Presence Fidelity Index (OPFI), the Operator Cognitive Load Index (OCLI), the Effective Command Latency (ECL), and the State Synchronization Accuracy (SSA).

We confirm that:

- This original work has not been published previously or considered elsewhere.
- All authors approved the final manuscript.
- The authors declare no conflicts of competing interest.

Generative AI Disclosure: In compliance with the journal’s guidelines on the use of generative Artificial Intelligence (AI), we disclose that an AI assistant was utilized during the writing and editing process for spelling, grammar verification, structural flow optimization, and formatting the manuscript’s bibliography and layout. No original data analysis, simulation results, or theoretical arguments were generated by the AI tool.

Thank you for your time, effort, and consideration of our work. We look forward to the peer review process.

Sincerely,

A handwritten signature in blue ink, appearing to read "J. Pandian".

Jason Pandian
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